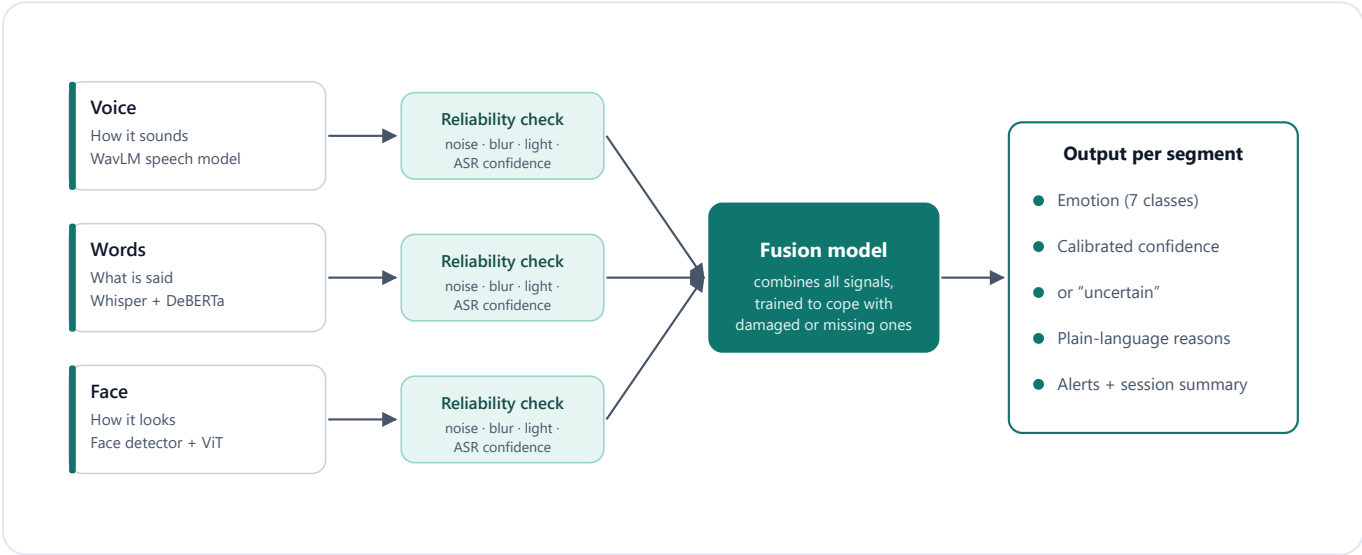
RAMEI
Emotion Intelligence

Reliability-Aware Adaptive Multimodal Emotion Intelligence

Recognising how people feel from **voice**, **words** and **face** together — and knowing when not to trust a signal.



DOCUMENT
Project presentation guide for the faculty review

PROJECT
AD-INFOCOMM internship

PRESENTED BY
[Student name] · [Roll number]

GUIDE
[Guide name]

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How to use this guide. Sections 1–11 explain the project; read them once so you can answer any question. Sections 12–14 are your preparation: the talk you will give, the demo you will run, and the questions you should expect. Every number in this guide comes from the project's measured results.

1 The project at a glance

RAMEI is an emotion-recognition system and web platform. It listens to the voice, reads the words and watches the face, checks how trustworthy each signal is at every moment, and combines them into one emotion with an honest confidence — or says "uncertain" instead of guessing.

0.769
Macro-F1, all three signals
unseen speakers (CREMA-D)

+17.7
points over voice alone
same speakers, same test

0.044
Calibration error (ECE)
target ≤ 0.05 — met

≤ 10%
Drop under every corruption
noise · blur · dark · occlusion

The problem

The starting point was a voice-only emotion recogniser that claimed about 70% accuracy. Tested fairly — on speakers it had never heard — it scored only **0.51**. It also failed whenever the voice was noisy and gave no sign of when it was wrong.

What we built

- Three AI branches: **voice** (WavLM), **words** (Whisper + DeBERTa), **face** (ViT).
- A **reliability score** for each signal, from measurable quality cues.
- A **fusion model** trained to keep working with damaged or missing signals.
- **Calibrated confidence**, an "uncertain" option, and plain-language reasons.
- A full **web platform**: upload, live camera + microphone, reports, alerts, export.

Technology

Layer	Tools
AI models	PyTorch · WavLM · Whisper · DeBERTa-v3 · ViT · MediaPipe
Backend	Python · FastAPI · WebSocket · SQLAlchemy
Frontend	React · TypeScript · Vite · Tailwind
Data	RAVDESS · CREMA-D (91 actors) · MELD
Delivery	Docker · nginx · GitHub Actions CI

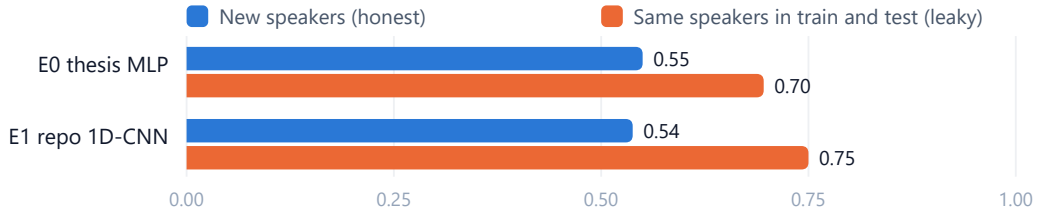
Status

All six milestones (M0–M5) are complete. The application runs end to end, in development and in Docker, with 64 automated tests passing on every change.

2 The problem we started from

The internship began from two audio-only projects: a 2021 thesis (MFCC features + MLP) and an open-source 1D-CNN that reported ~70% accuracy. Before improving anything, we measured them honestly.

Same models, same data — only the evaluation split differs (macro-F1)



RAVDESS, single split. The original ~70% claim only appears when the test speakers were also in training.

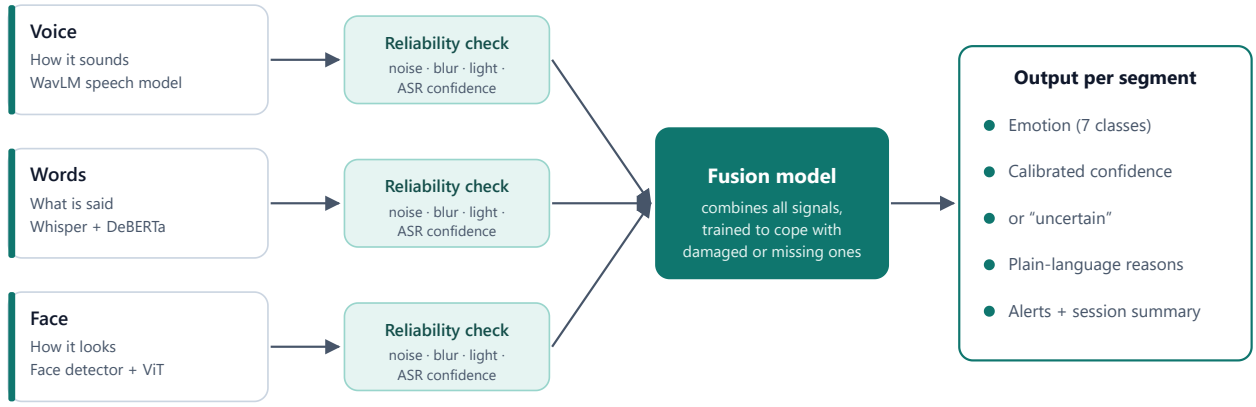
Why the 70% was misleading. Emotion datasets contain many clips from each actor. If the same actor appears in training and test data, the model can partly recognise the *person* instead of the *emotion*. This "speaker leakage" added 15–21 points. Every result in this project uses **speaker-independent** testing: the model is always tested on people it has never heard or seen.

Gaps in the baseline

#	Gap	Consequence
G1	Voice is the only signal	Fails when the voice is unclear, flat or missing
G2	No idea of signal quality	Confidently wrong on noisy or silent input
G3	Uncalibrated confidence	A score of 90% does not mean 90% likely
G4	Breaks without inputs	Nothing works if the mic or camera is off
G5	One label per file	No timeline, no alerts, no insight
G6	Unclear evaluation	Results do not transfer to new people
G7	Notebook code only	Not usable by anyone

3 Our idea

People show emotion through several channels at once. When one channel is unreliable — a noisy room, a dark camera, mumbled words — a person naturally relies on the others. RAMEI is built to do the same.



Each signal is analysed separately, checked for quality, and combined into one decision.

1 · Measure quality

Signal-to-noise ratio for the voice; speech-recognition confidence for the words; face visibility, blur and lighting for the face.

2 · Combine robustly

A fusion model trained on deliberately damaged and missing inputs, so a weak signal does not drag the decision down.

3 · Be honest

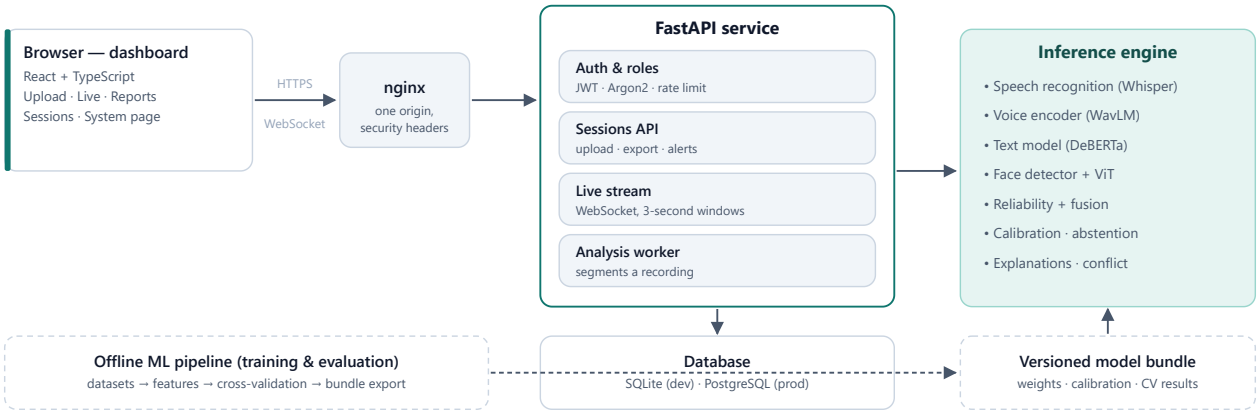
Confidence is calibrated; low-confidence segments are marked "uncertain"; every result comes with the reasons behind it.

Before and after

	Original baseline	RAMEI
Signals	Voice only	Voice + words + face
Fair accuracy (unseen people)	0.51 macro-F1 (RAVDESS)	0.769 macro-F1 (CREMA-D)
Noisy room	Not handled	0.715 — only a 7% drop
Confidence	Uncalibrated (ECE 0.28)	Calibrated (ECE 0.044)
When unsure	Guesses anyway	Says "uncertain"
Explanations	None	Per-segment reasons
Product	Notebook	Web platform with live analysis

4 System architecture

RAMEI has two halves that meet at one contract. An **offline ML pipeline** trains and evaluates the models and exports a versioned **model bundle**. The **online platform** loads that bundle and serves users.



The browser talks to one address; nginx routes pages and the API, including the live WebSocket.

Main components

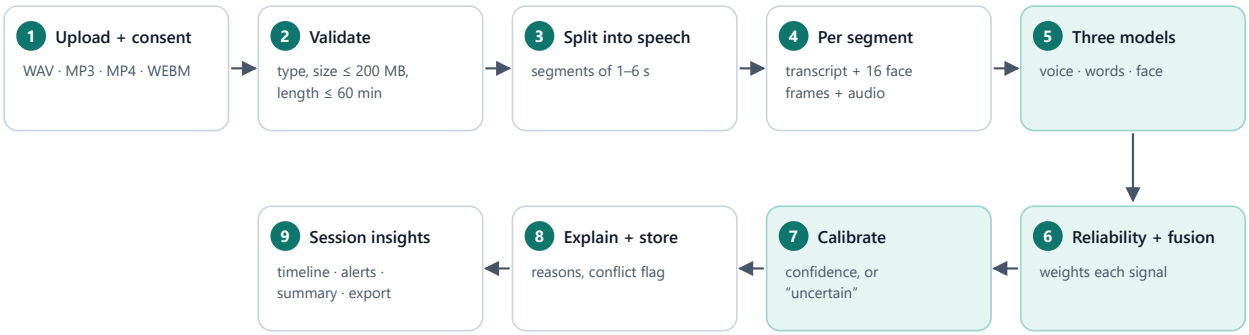
Component	Responsibility
Dashboard (React)	Upload, live analysis, session reports, system page
API (FastAPI)	Login, roles, sessions, uploads, export, alerts
Live stream	Receives audio and frames, returns results every 3 s
Inference engine	Runs the models, fusion, calibration, explanations
Model bundle	Weights + calibration + measured results, versioned
Database	Sessions, segments, alerts (no raw media by default)

Design decisions

- **One versioned bundle** — the platform never loads loose training files, so every result records exactly which model produced it.
- **Same interfaces, simpler parts** — SQLite instead of PostgreSQL and in-process workers instead of a queue for the internship build (ADR-0008); both can be swapped without changing the API.
- **Privacy by default** — consent before any capture; recordings deleted after analysis.
- **Decisions are written down** — architecture decision records (ADRs) explain every major choice.

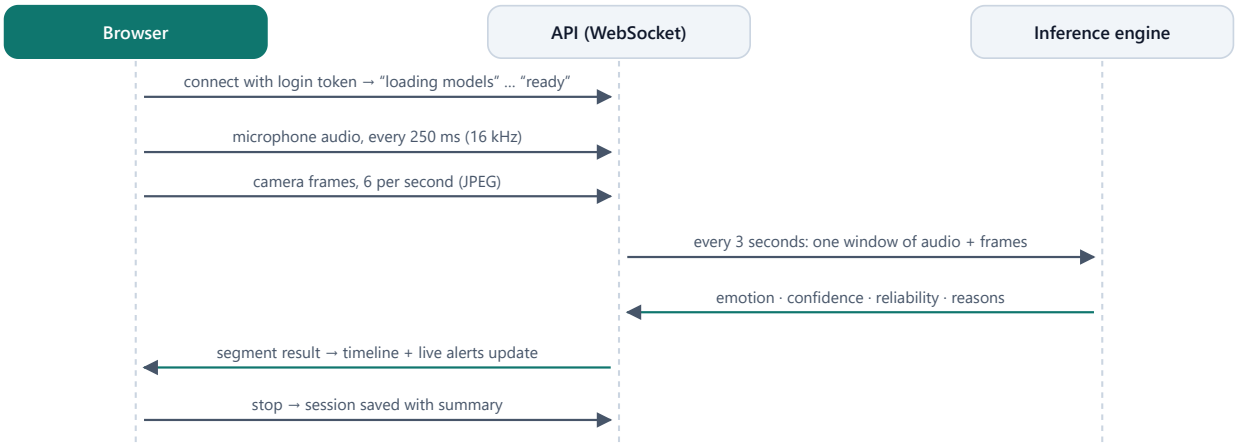
5 How an analysis works

Upload mode — a recorded call or video



Raw recordings are deleted after analysis by default; only results are stored.

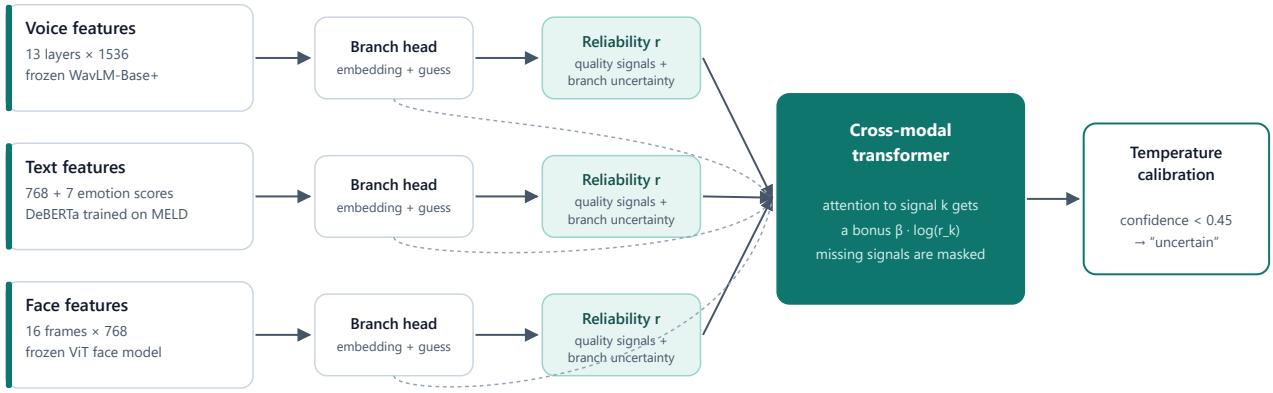
Live mode — camera and microphone in the browser



If analysis falls behind, video frames are dropped first so audio stays real-time.

6 The AI model inside

Large pretrained models turn each signal into numbers (“features”). RAMEI's own fusion model then learns how to combine them. Training the fusion model on stored features is what made a 4 GB laptop GPU sufficient.



Each branch makes its own guess; reliability scores and a transformer combine them; the result is calibrated.

During training every example is randomly damaged, so the model learns to cope:

Clean original input	**Noise 10 dB** busy office	**Noise 0 dB** noisy room, ASR re-run	**Blur** out-of-focus face
Dark poor lighting	**Occluded** half the face hidden	**Word drop** 50% of words lost	**Missing** camera or mic off

Each signal is clean 40% of the time; each is dropped entirely 20% of the time (at least one always remains).

Why “frozen” encoders?

On 24 RAVDESS speakers, fine-tuning the voice encoder (LoRA) and keeping it frozen scored the same (0.686 vs 0.701, within noise). Frozen is cheaper and more stable, so the fusion model uses frozen features.

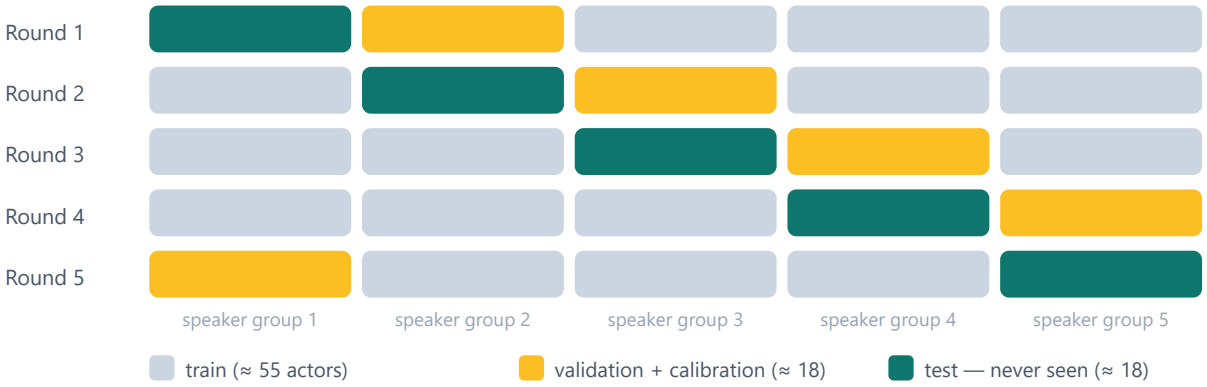
How confidence is calibrated

A single “temperature” is fitted on validation speakers so that, for example, 80% confidence really means about 80% correct. Below 0.45 the segment is “uncertain”.

7 How we evaluated it

Dataset	What it contains	Used for
RAVDESS	24 actors (12 F / 12 M), 1,440 acted speech clips	Baselines and the voice branch
CREMA-D	91 actors, 7,442 clips with audio and video	Face branch and the full fusion model
MELD	~13,000 lines of TV dialogue (Friends)	The words branch (natural conversation)

5-fold speaker-independent cross-validation on CREMA-D (91 actors)



Every actor is tested exactly once, on a model that never heard or saw them. Scores are the mean ± spread over the 5 rounds.

What we measure

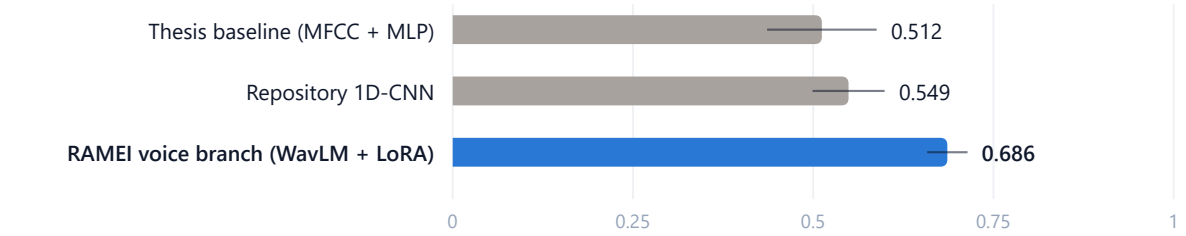
Metric	In plain words
Macro-F1	Accuracy that treats every emotion equally, so rare emotions count as much as common ones
ECE (calibration error)	How far stated confidence is from real accuracy — lower is better
Relative drop	How much the score falls when a signal is damaged or missing
Selective accuracy	Accuracy on the segments the model chose to label
Per-gender F1	Whether it works equally well for female and male speakers

CREMA-D has six emotions (no “surprise”), so its scores are over six classes. Comparisons within CREMA-D are like-for-like; comparisons with the seven-class RAVDESS numbers are indicative.

8 Results

Step 1 — a stronger voice model

Voice only, unseen speakers — macro-F1 (RAVDESS, 6-fold cross-validation)

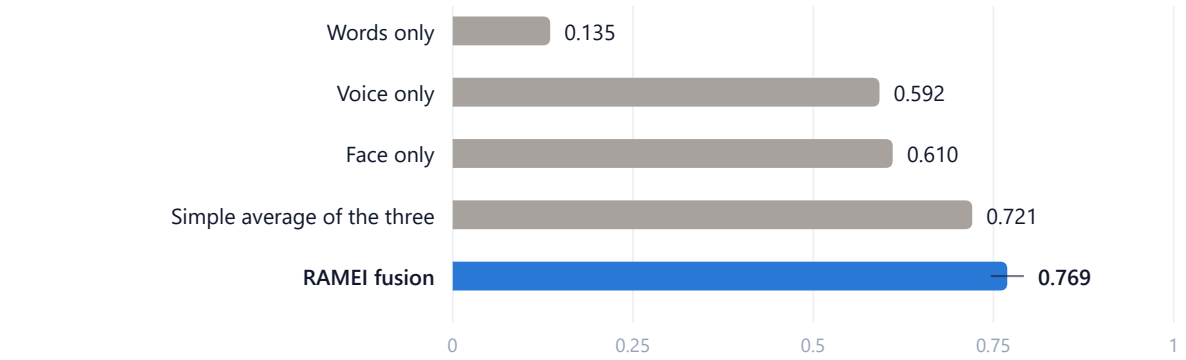


Bars show the mean; the thin line shows ± one standard deviation across folds.

The WavLM voice branch alone improves the honest baseline by **+17 points**, which already meets the "baseline + 10 points" target.

Step 2 — combining voice, words and face

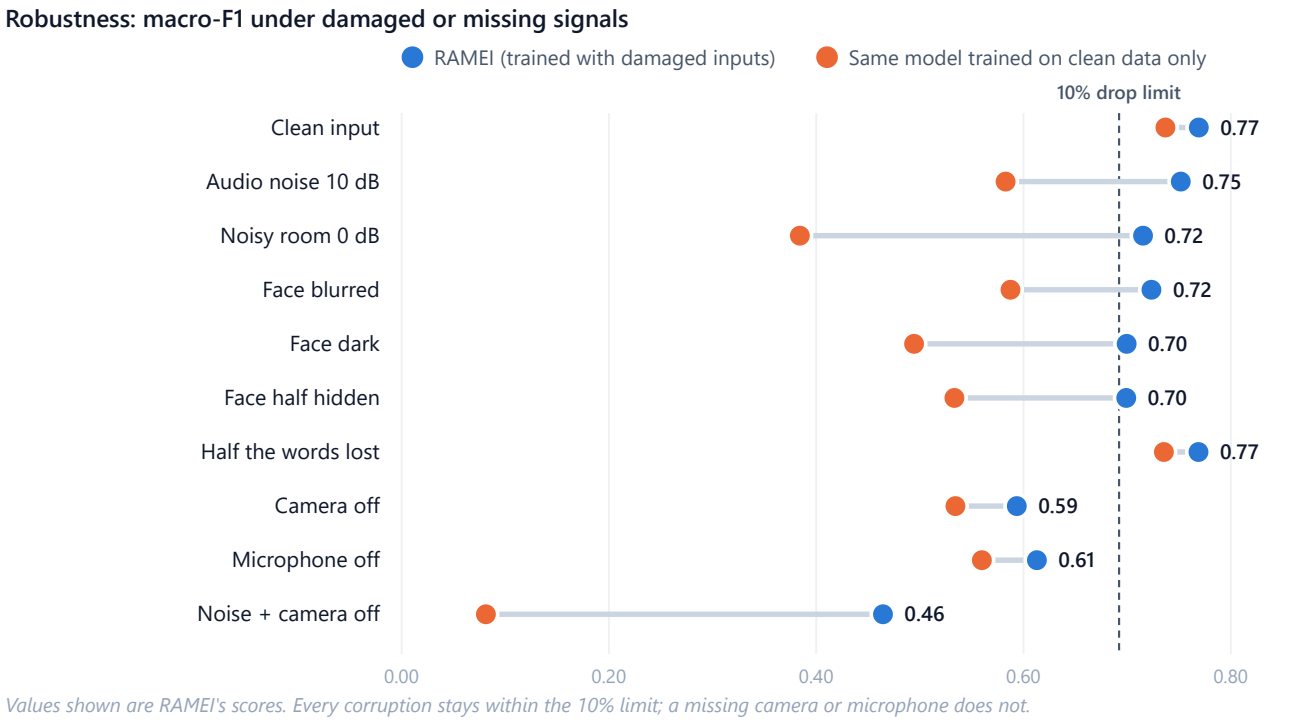
All three signals, unseen speakers — macro-F1 (CREMA-D, 5-fold cross-validation)



Words score low here because every CREMA-D actor reads the same 12 neutral sentences.

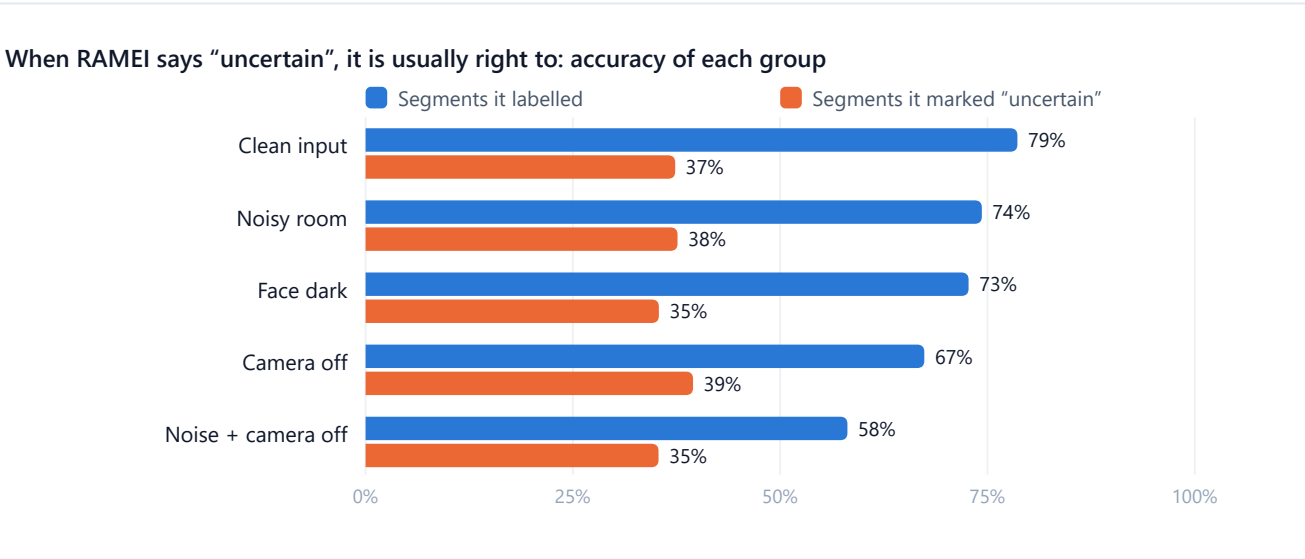
Fusion scores **0.769**: **+17.7** points above voice alone and **+15.9** above the best single signal (face), on the same unseen speakers. The learned fusion is also **+4.9** points better than simply averaging the three.

Step 3 — staying reliable when signals are damaged



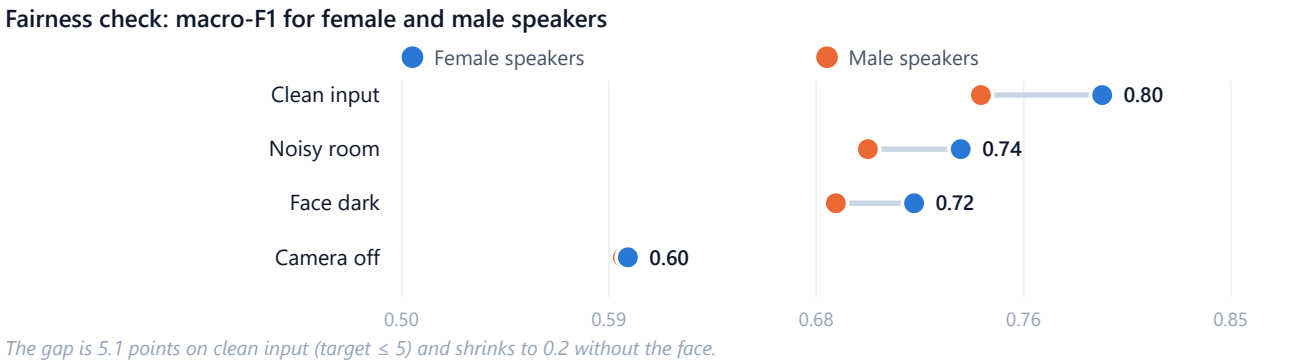
Every corruption — noise, blur, darkness, occlusion, lost words — keeps RAMEI within 10% of its clean score. The same model trained only on clean data collapses in a noisy room (0.38). With a whole signal missing, RAMEI falls back to roughly the level of the remaining signal: graceful, not catastrophic.

Step 4 — knowing when it does not know



On clean input RAMEI labels 96% of segments with **79%** accuracy; the 4% it marks "uncertain" are only 37% accurate — exactly the ones worth not trusting. As conditions worsen it abstains more (4% → 28% → 46%) instead of guessing.

Step 5 — fairness



The gap is 5.1 points on clean input (target ≤ 5) and shrinks to 0.2 without the face.

Requirement scorecard

Target	Requirement	Result	
NFR-03	Fused score ≥ audio baseline + 10 points	+17.7 points	**Met**
NFR-04	≤ 10% drop with a degraded signal	2–9% across all corruptions	**Met**
NFR-04	≤ 10% drop with a missing signal	20–23% (falls to one-signal level)	**Not met**
NFR-05	Calibration error ECE ≤ 0.05	0.044 clean · 0.043–0.057 degraded	**Met** (clean)
NFR-09	Gender gap ≤ 5 points	5.1 clean · 0.2–3.9 degraded	**Borderline**
FR-14	Abstain when uncertain	37% vs 79% accuracy split	**Met**

Speed

Setting	Measured
Analyse a 17-second three-signal video (laptop GPU, RTX 3050)	4.2 s — about 4× faster than real time
Same video inside Docker on CPU only	14.3 s — still faster than real time
Web API, 10 concurrent users	105 requests/s, median 48 ms, 0 errors

9 What we learned — honestly

Faculty value honesty over hype. These findings are reported exactly as measured.

Finding 1 · The reliability bias did not add accuracy

We removed the reliability term from the fusion attention (experiment E9) and the score did not change (0.766 vs 0.769). The reliability *scores* still work — they fall when a signal is damaged — and they drive explanations, abstention and conflict detection.

Finding 2 · Training on damaged data is what matters

The same model trained only on clean data (E10) collapsed under noise: 0.38 vs 0.72 in a noisy room, 0.08 vs 0.46 with noise and no camera. Robustness came from *how* the model was trained.

Finding 3 · The face dominates on acted data

On CREMA-D the fusion gives the face about 84% of the weight on average, because acted faces are frontal and well lit. On natural conversations the balance would differ — a reason to test on real recordings next.

Finding 4 · Honest evaluation changes the story

The baseline's 70% became 51% under a fair test. Every number in this project is speaker-independent and averaged over several rounds.

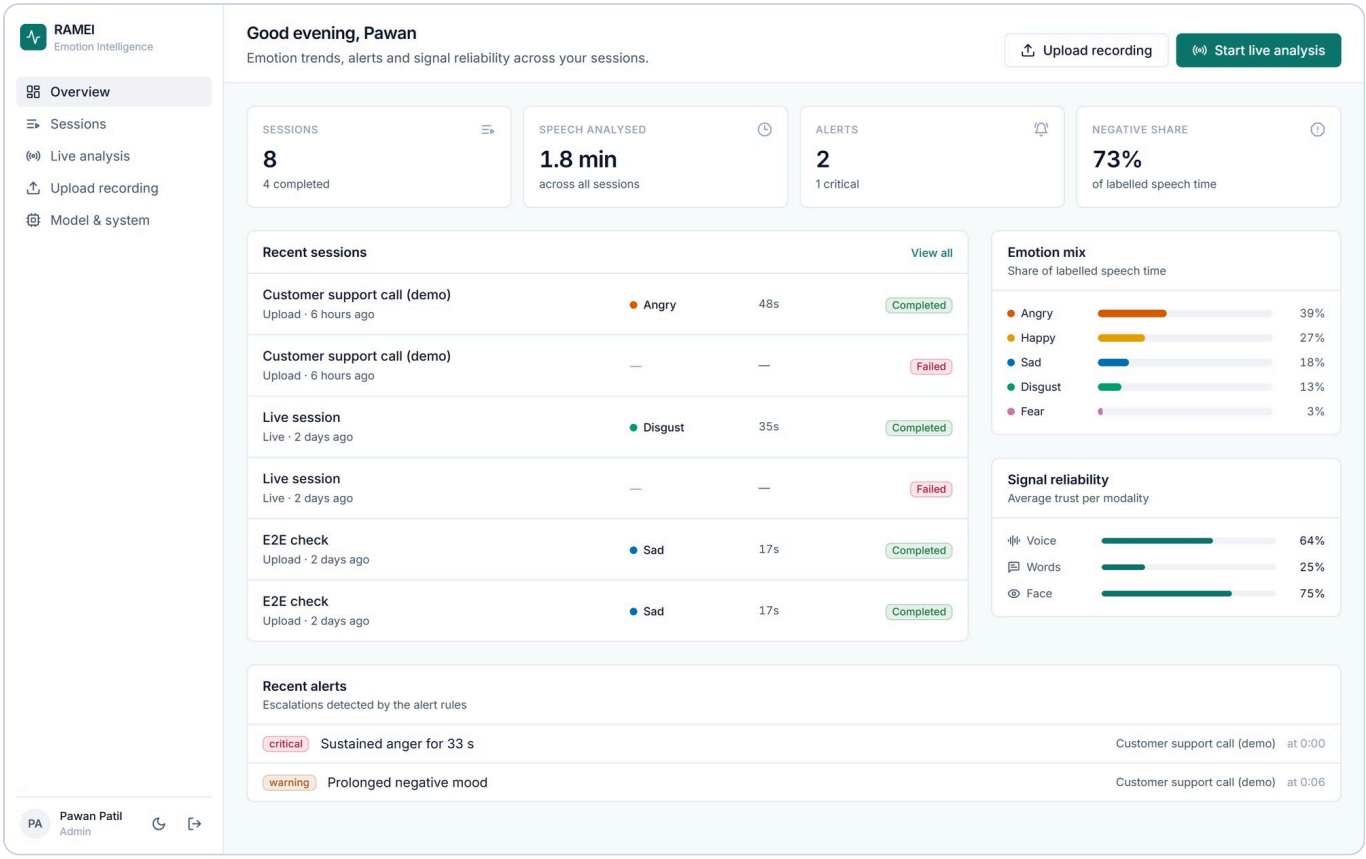
Limitations

- **Acted datasets.** Actors read fixed sentences, so the words carry little emotion and real calls will differ. Licensed conversational data (IEMOCAP, MSP-Podcast) is the next step.
- **A missing signal costs 20–23 points**, back to single-signal level; calibration also weakens then.
- **Valence/arousal** (positive–negative, calm–excited) is estimated from class probabilities, not trained.
- **Prototype deployment.** Runs on one machine (SQLite, CPU in Docker); production would add PostgreSQL, HTTPS and a GPU server.

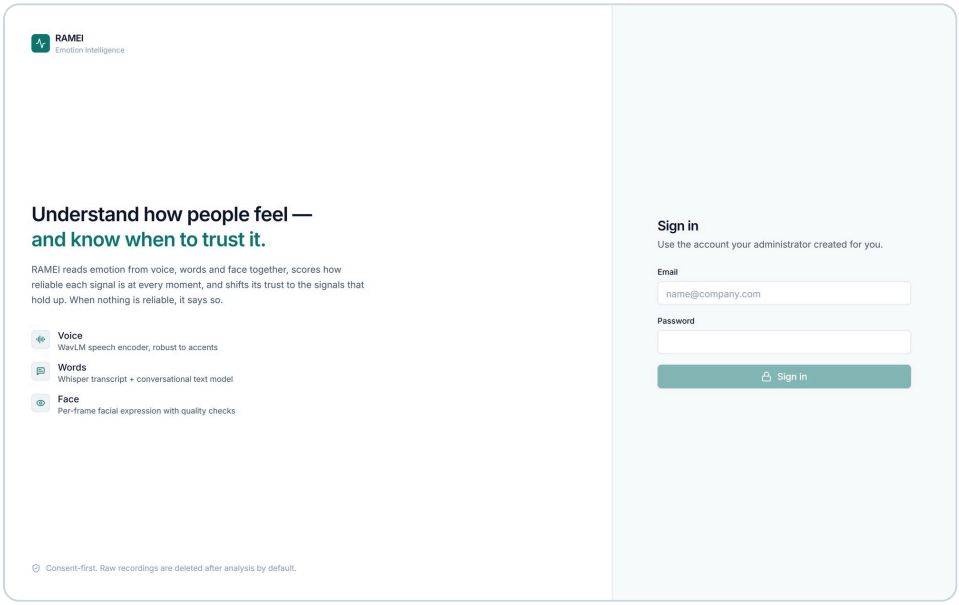
Ethics

Emotion recognition is probabilistic and culturally variable. RAMEI must not be used for decisions about individuals (hiring, discipline, diagnosis) or for monitoring people without consent. The platform requires consent before every analysis, deletes recordings by default, abstains when unsure and explains every result.

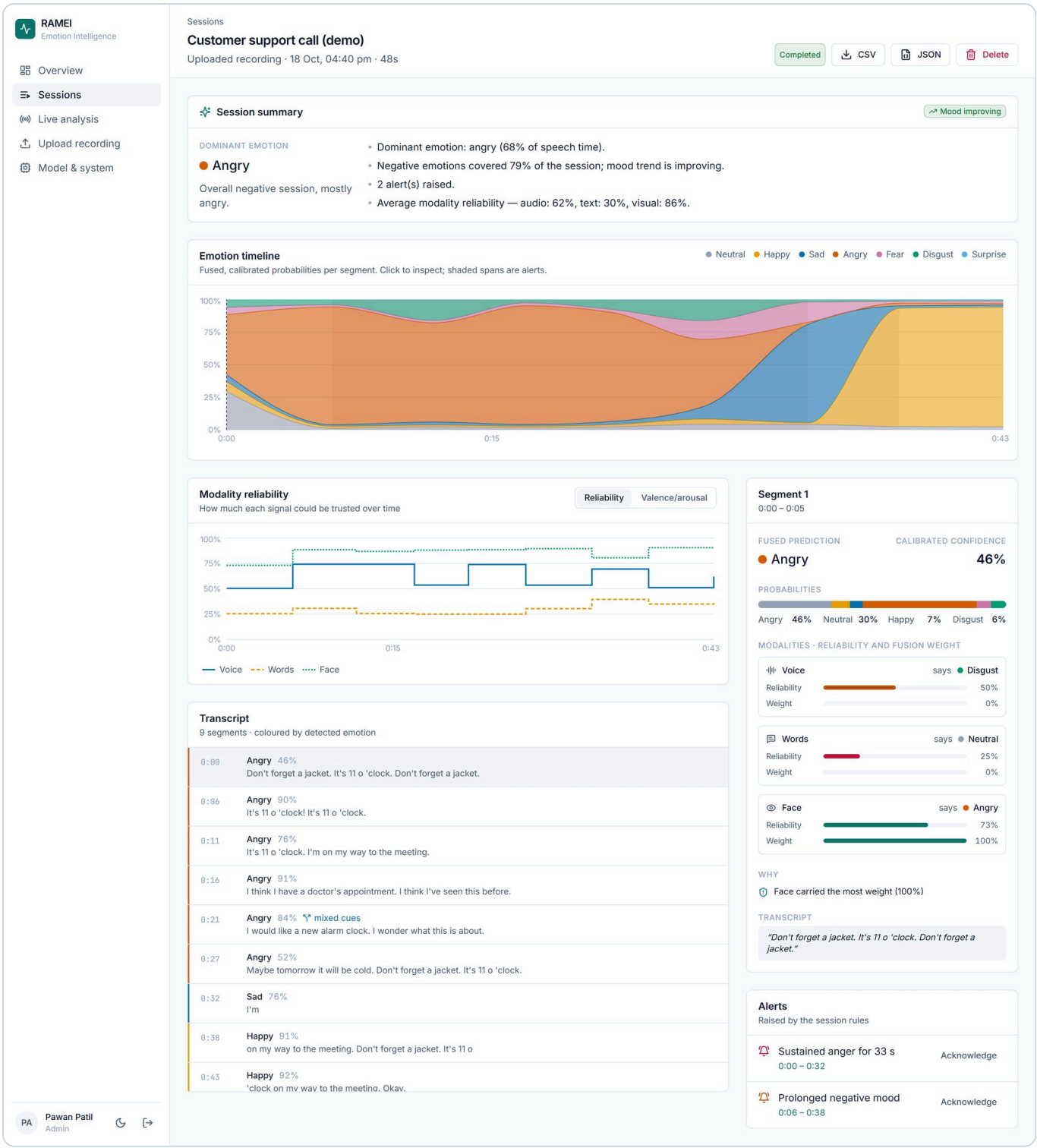
10 The application



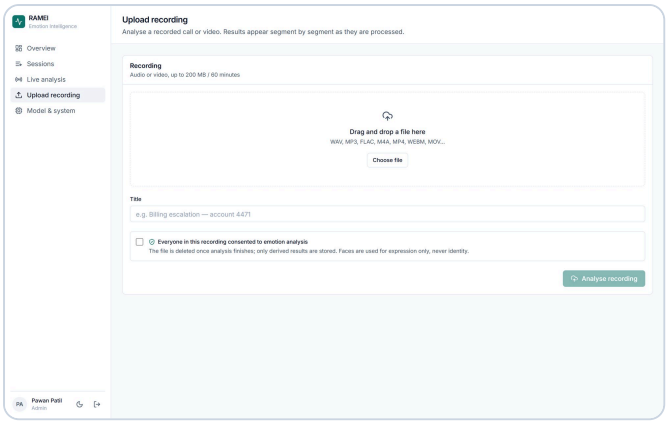
Overview — sessions, emotion mix, signal reliability and recent alerts across the organisation.



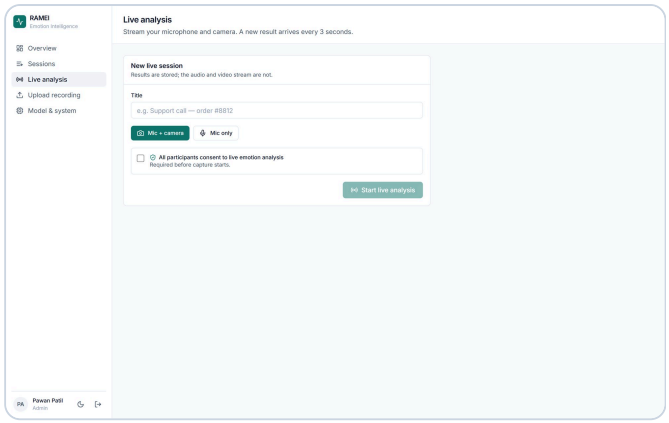
Sign-in — accounts are created by an administrator; there is no public sign-up.



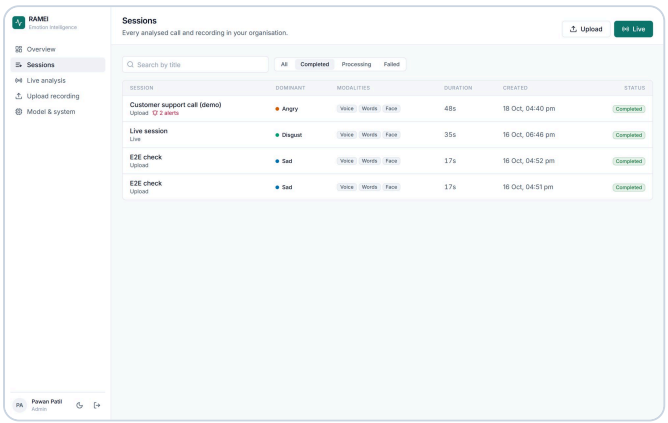
Session report for a 48-second demo call: summary, emotion timeline (shaded = alert), signal reliability over time, colour-coded transcript, and the segment inspector with probabilities, per-signal reliability and weight, and the reasons. Alerts fired for sustained anger (33 s) and prolonged negative mood.



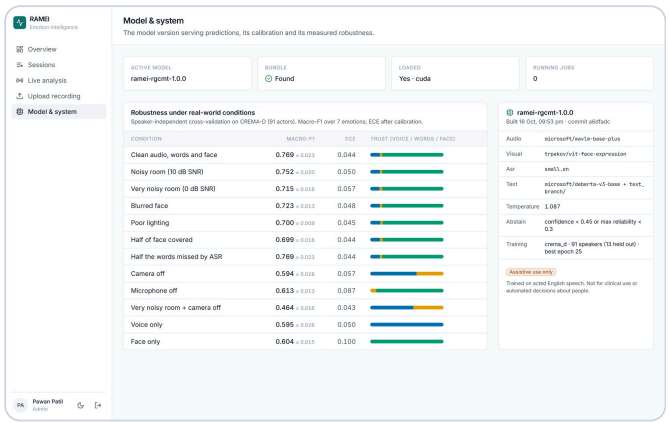
Upload — any WAV, MP3, MP4 or WEBM up to 200 MB; consent is required.



Live analysis — microphone + camera or microphone only; results every 3 seconds.

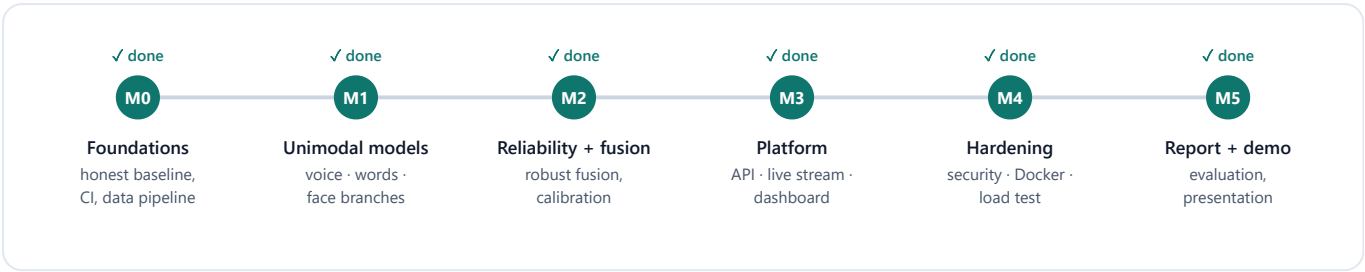


Sessions — search and filter every analysis; export to CSV or JSON.



Model & system — the active model, its calibration and its measured robustness.

11 Engineering quality



Quality and testing

Area	Evidence
Automated tests	64 tests: 39 ML, 20 backend, 5 frontend
Continuous integration	GitHub Actions: lint, type-check, tests, build on every push
End-to-end check	Real video through API and dashboard, GPU and Docker
Load test	0 errors at 1, 10 and 50 concurrent users
Documentation	PRD, TRD, architecture, 2 ADRs, 3 milestone reports, final report

Security and privacy

Control	How
Login	JWT tokens, Argon2 password hashing
Access	Agent / analyst / admin roles; organisations isolated
Abuse	Rate-limited login attempts
Uploads	Random file names, size cap, content validated
Browser	Strict content-security policy and security headers
Privacy	Consent per session; recordings deleted after analysis

Deployment

The whole platform runs with one command — `docker compose up` — as two containers (dashboard and API) behind nginx, with the database and model cache in persistent volumes. Building and running it in Docker found and fixed six deployment bugs that the developer setup had hidden.

12 Presentation script — about 12 minutes

Use one slide per row. Each slide can reuse a figure from this guide. Speak to the faculty, not the screen; the "say" boxes are suggested phrasings, not text to read aloud.

Time	Slide	What to cover
0:00	**Title**	Introduce yourself, the internship and the one-line idea: emotion from voice, words and face — that knows when not to trust a signal.
0:45	**The problem**	Baseline: voice only, claimed ~70%. Show the leakage chart: on new speakers it is 51%. Explain speaker leakage in one sentence.
2:00	**Our idea**	Show the concept diagram. Three signals, a quality check for each, one combined, honest answer.
3:00	**Architecture**	Offline pipeline produces a versioned model bundle; the platform serves it. Point to the browser, API, engine.
4:00	**How it works**	Walk the nine-step upload flow; mention live mode returns a result every 3 seconds.
5:00	**The model**	Branch heads, reliability scores, transformer, calibration. Mention training on damaged inputs.
6:00	**Evaluation**	Three datasets; every actor tested on a model that never saw them (cross-validation diagram).
7:00	**Results**	0.51 → 0.69 (voice) → 0.769 (fusion). Robustness chart: within 10% under every corruption.
8:15	**Honesty**	Abstention: 79% vs 37%. Then the honest finding: robustness came from training, not the reliability bias.
9:15	**Live demo**	Run the demo (Section 13): show a report, the inspector, an alert, then live mode.
11:15	**Close**	Scorecard, limitations, next steps (real conversational data). Thank the faculty and invite questions.

> **SAY** *"The original model claimed 70%. When we tested it fairly — on people it had never heard — it scored 51%. That honest number is what we set out to beat."*

> **SAY** *"RAMEI looks at three signals, checks how trustworthy each one is, and when it cannot be confident it says so instead of guessing."*

> **SAY** *"On 91 actors it never saw during training, RAMEI scores 0.77, almost 18 points above voice alone, and it stays within 10% when the audio is noisy or the face is dark."*

> **SAY** *"We also report what did not work: removing the reliability term did not change accuracy. Robustness came from training on damaged inputs."*

13 Live demo runbook

The day before

1. Record your own 20–40 second clip: face well lit, a clear emotion (for example annoyed, then calm).
2. Create degraded copies: `python scripts/make_demo_clips.py my_clip.mp4`
3. Upload the clean, noisy and dark copies once, so the reports are ready.
4. Charge and plug in the laptop; close other GPU apps; silence notifications.
5. Rehearse the full talk once with a timer.

Thirty minutes before

1. Start the backend: `uvicorn app.main:app --port 8000` (in `backend/`).
2. Start the dashboard: `npm run dev` (in `frontend/`) and open `localhost:5173`.
3. Sign in, open one report so the models are loaded and warm.
4. Allow camera and microphone for `localhost`; zoom the browser to 110–125%.

During the demo (about 2 minutes)

1. **Overview** — point at the emotion mix, reliability and the alert list.
2. **Open the demo call report** — read the summary; show the timeline with the shaded anger alert.
3. **Click a segment** — show probabilities, each signal's reliability and weight, and the "why" lines.
4. **Compare clean vs noisy upload** — voice reliability falls, the reason says why, the emotion holds.
5. **Live analysis** — start with camera; speak for 15 seconds; cover the lens: the face becomes unavailable and voice + words carry the decision.
6. **Model & system** — the measured robustness table, for anyone who asks "how do you know?"

Backup plan

- If live capture fails, show the pre-computed reports — they prove the same points.
- If the laptop fails, use the screenshots in Section 10 of this guide.
- Docker alternative: `docker compose up -d`, then `localhost:8080` (CPU, slower).

14 Questions faculty may ask

Why is your accuracy lower than the original 70%?

Because we test on people the model has never seen. The 70% came from the same actors appearing in training and test; under a fair test the original scores 51%. Our 0.769 is on unseen speakers.

What is new compared to existing work?

The combination: three signals, a measured reliability for each, a fusion model trained on damaged and missing inputs, calibrated confidence with an abstain option, explanations, and a working platform with live analysis — all evaluated speaker-independently.

Does the reliability module actually help?

The reliability scores track damage correctly and drive explanations, abstention and conflict detection. But the reliability term inside the fusion did not add accuracy (ablation E9 equals E8). The robustness came from training with damaged inputs — we report this openly.

How do you handle a missing camera or microphone?

Missing signals are masked out of the fusion, so the model uses whatever is available. The score falls to about the level of the remaining signal (0.59–0.61), rather than failing.

What does "uncertain" mean?

The calibrated confidence is below 0.45. Those segments are only about 37% accurate, so refusing to label them is the right behaviour.

Why not train everything end to end?

Our GPU has 4 GB. Fine-tuning the voice encoder matched the frozen encoder on 24 speakers, so frozen features were cheaper and equally good.

Is it biased?

We measured it: female 0.796 vs male 0.745 on clean input — a 5.1-point gap, just above our 5-point target, and almost zero when the face is removed. The face model is the likely source.

Will it work on real phone calls?

Not proven yet. The data is acted. The next step is licensed conversational data such as IEMOCAP or MSP-Podcast.

How fast is it?

About 4× faster than real time on the laptop GPU, and still faster than real time on CPU inside Docker. Live mode returns a result every 3 seconds.

What about privacy?

Consent is required before any capture; recordings are deleted after analysis by default; only results are stored; accounts are created by an administrator.

Why these three datasets?

RAVDESS for a fair comparison with the baseline, CREMA-D because it has video and 91 actors for strong speaker-independent testing, and MELD because it is natural dialogue for the text model.

How do you know the numbers are right?

Five-fold cross-validation over 91 speakers, results stored as data files, an analysis script that reproduces the pooled score exactly, and CI running 64 tests on every change.

What would you do with more time?

Test on conversational data, train a real valence/arousal head, fine-tune encoders end to end with more GPU memory, and move to PostgreSQL and GPU serving for production.

Could this be misused?

Yes, which is why we state it must not be used for decisions about individuals or monitoring without consent, and why the system defaults to consent, deletion, abstention and explanation.

15 Glossary

Multimodal
Using more than one kind of input — here voice, words and face.

Modality / signal
One input channel (voice, words or face).

Encoder
A large pretrained model that turns raw audio, text or images into numbers (features).

Frozen
An encoder used as-is, without retraining its weights.

LoRA
A cheap way to fine-tune a large model by training a small number of extra weights.

Fusion
Combining the three signals into one decision.

Reliability score
RAMEI's estimate (0–1) of how trustworthy a signal is for a given segment.

Transformer / attention
A neural network that decides how much each input should influence the output.

Macro-F1
An accuracy measure that treats every emotion equally (1.0 = perfect).

Calibration / ECE
Whether stated confidence matches real accuracy; ECE is the average gap (0 = perfect).

Temperature scaling
A single learned number that adjusts confidence so it becomes calibrated.

Abstention
Deliberately answering "uncertain" instead of guessing.

Speaker leakage
When test speakers also appear in training, inflating results.

Cross-validation
Repeating training and testing on different groups of people, then averaging.

Ablation
Removing one part of the system to measure what it contributes.

SNR (dB)
Signal-to-noise ratio: how loud speech is compared to background noise; 0 dB means equally loud.

ASR
Automatic speech recognition — turning speech into text (Whisper).

WebSocket
A connection that keeps data flowing both ways in real time, used for live mode.

Docker
Packaging the application with everything it needs, so it runs the same on any machine.

Where to find more. Full details are in the repository: `docs/reports/Final-Report.md` (and the Word version), the milestone reports `M0`, `M1` and `M2`, the architecture document, and the decision records in `docs/adr/`.